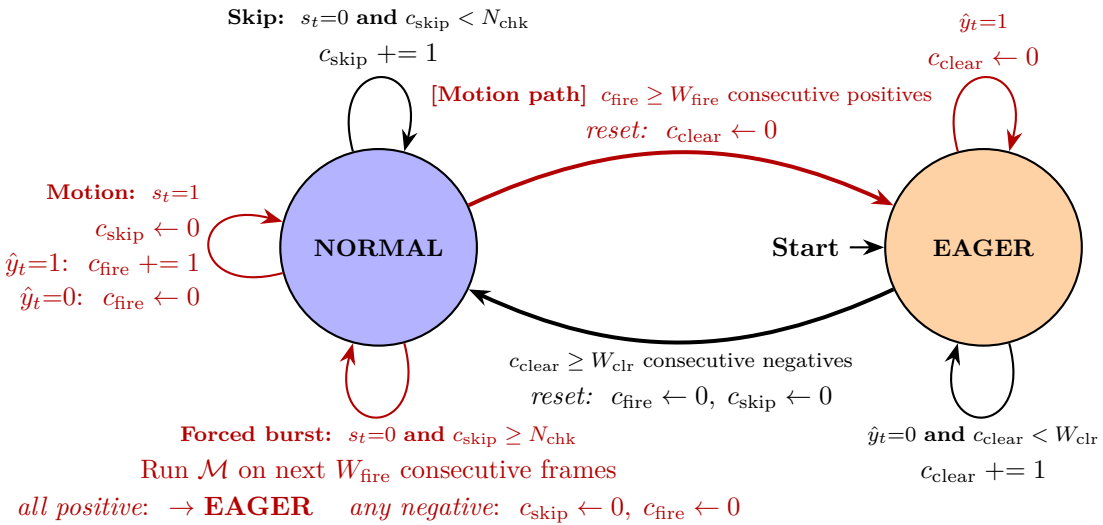




Modes

- **NORMAL** $\mathcal{S}$ (motion sensor) active. Frames skipped unless motion detected or silence limit N_{chk} is reached.
- **EAGER** $\mathcal{S}$ bypassed. $\mathcal{M}$ (fire model) runs on **every** frame. *Each video starts in this mode.*

Arrow colours

- $\mathcal{M}(f_t)$ **is invoked** on this transition
- $\mathcal{M}(f_t)$ **not invoked** (frame skipped or counter only)

Symbols

- $s_t \in \{0, 1\}$ $\mathcal{S}$ output: 0=no motion, 1=motion
- $\hat{y}_t \in \{0, 1\}$ $\mathcal{M}$ output: 1=fire/smoke, 0=safe
- c_{fire} Consecutive positive $\mathcal{M}$ results on motion path; reset on any negative or T1 exit
- c_{clear} Consecutive negative $\mathcal{M}$ results in EAGER; reset on any positive or T2 entry
- c_{skip} Consecutive skipped frames in NORMAL; reset on any $\mathcal{M}$ call
- W_{fire} (1) Motion streak length to enter EAGER; (2) Burst size for forced check
- W_{clr} Consecutive negatives in EAGER needed to exit
- N_{chk} Max consecutive skipped frames before forced burst